SDG BLOCKCHAIN ACCELERATOR

Plastiks x El Salvador: Blockchain-Powered Waste Traceability for Sustainable Development

This document outlines the market penetration strategy, branding approach, user acquisition plan, go-to-market strategy, and long-term scaling roadmap for the Plastiks x UNDP El Salvador pilot developed within the SDG Blockchain Accelerator program.

1. Introduction and Problem Framing

El Salvador faces a critical plastic pollution problem, with packaging and single-use plastics accumulating in landfills, waterways, and coastal areas. A new national waste law and the creation of ANDRES (Autoridad Nacional de Residuos Sólidos) set the stage for Extended Producer Responsibility (EPR), but the country lacks a unified system to track and verify plastic recovery. Currently, data on collected and recycled plastics is fragmented, unreliable, and often informal, making it difficult for regulators to enforce EPR laws or for producers to demonstrate compliance. Informal recyclers and cooperatives – many of them women and low-income workers – struggle without formal recognition or transparent records of their work.

This pilot addresses these challenges by deploying a blockchain-enabled traceability platform designed to empower grassroots recyclers, integrate informal waste workers, and create transparent value for recovered plastic. It turns everyday plastic recovery into compliance-grade, auditable data. Each kilogram of plastic collected can be recorded with evidence (such as weight, GPS, or photo proof) and anchored on a public blockchain. Verified recovery actions are then tokenized as digital certificates or "plastic credits," enabling producers to substantiate their recycling and regulators to oversee performance. The result is a win-win: recyclers earn fair, prompt payments; brands gain credible sustainability claims; and El Salvador gains a robust, data-driven foundation for its circular economy.

This narrative proposal, prepared in collaboration with UNDP El Salvador and Cardano's SDG Accelerator framework, outlines how this solution transforms a national challenge into an opportunity. It demonstrates how innovative technology, strong partnerships, and a people-centered design can align with UNDP's climate goals and Cardano's vision of inclusive, low-carbon blockchain adoption.

2. Challenge and Opportunity

El Salvador's challenge is emblematic of many emerging economies. Despite legal mandates for waste management and producer responsibility, practical implementation remains weak. Waste collection is often manual and uncoordinated,

while records of recycling are kept in silos or not at all. As a result, producers cannot verify how much of their plastic packaging is actually recycled, governments cannot track progress towards reduction targets, and communities lack the incentives and support to reduce pollution. Informal waste pickers, who do much of the actual collection, are economically marginalized and rarely included in official systems. This leads to lost opportunities: valuable plastic is wasted in landfills or waterways, the informal sector misses out on income, and El Salvador's environmental and climate commitments fall short.

Against this backdrop lies a powerful opportunity. Globally, Extended Producer Responsibility (EPR) and Deposit Return Schemes (DRS) are emerging as the most effective tools to close material loops and combat climate change through recycling. Blockchain technology brings a new dimension to these tools by offering transparency, trust, and automation at scale. Cardano's low-energy, proof-of-stake blockchain provides an energy-efficient backbone for public, verifiable records. By digitizing every step of the plastic recovery process, we can create trusted "proofs of recycling" that satisfy regulators, meet international ESG reporting standards, and even unlock new climate finance.

In El Salvador, these trends converge. A strong legal framework (such as the Special Waste Law) exists on paper, and UNDP has mobilized key stakeholders (ANDRES, FUNDEMAS, municipal agencies, and private companies) to engage. The plastic waste crisis thus becomes an accelerator for innovation: we can leapfrog traditional systems by building a cutting-edge digital MRV (Measurement, Reporting, Verification) platform. This pilot also taps into development goals: improving livelihoods for waste workers (SDG 1, 8), ensuring inclusive institutions (SDG 16), and fostering global partnerships (SDG 17). In short, the challenge of insufficient plastic recovery data transforms into the opportunity to pioneer a scalable, blockchain-based compliance layer, positioning El Salvador as a leader in the sustainable circular economy.

3. Market Segmentation and User Profiles

The solution serves a diverse ecosystem of stakeholders, each with distinct needs and roles:

- **Grassroots Recyclers and Waste Collectors:** These are the men and women who physically collect, sort, and transport plastic waste – often in cooperatives or informal groups. They face low margins and lack formal recognition. By using the platform, recyclers can **log recovery activities with evidence**, gain instant, transparent compensation (in ADA or local currency), and build a verified track record of their work. Training and easy interfaces (including feature-phone compatibility and offline modes) ensure even remote or low-literacy workers can participate. Over time, their income security, working conditions, and social status improve through accreditation and inclusion in the formal value chain.
- **Municipal and Inter-Municipal Waste Authorities:** Cities and local governments (often under the new ANDRES framework) manage waste collection but struggle with fragmented data. This platform provides a single digital dashboard aggregating all recovery data across regions and materials. Municipalities can monitor performance against targets, plan investments (like new recycling facilities), and coordinate with national regulators. By standardizing data capture (weight, geolocation, etc.), even different collection agencies speak the same "data language," reducing inefficiencies and disputes.
- **Producers and Brands (EPR-Obligated Companies):** Beverage companies, packaging manufacturers, and retailers are legally required to ensure their packaging is recycled. They need credible evidence to satisfy regulators and to claim recycled content for sustainability reporting. As key users of the platform, they receive **verifiable reports on collected volumes tied to their products (e.g. PET bottle SKUs)**. Early participants ("private anchors") might co-finance recovery pilots to jumpstart the system. The platform enhances their brand image (demonstrating genuine environmental action) and can generate marketable "plastic credits" or offset certificates, creating an economic incentive to invest in recycling.
- **Extended Producer Responsibility Organizations (PROs) and NGOs:** Entities like FUNDEMAS or local recycling federations that coordinate collection under

EPR frameworks will use the platform to streamline operations. For example, a PRO managing a specific industry can integrate member data, automate reporting, and distribute incentives via smart contracts. NGOs working on waste management gain a powerful tool for monitoring projects and advocating for policy, as data is centrally accessible.

- **Retailers and Deposit Return Operators:** In a deposit return scenario, retail chains and bottlers collect returned bottles. They can record DRS events on the platform (aggregated returns by SKU and location) and reconcile deposits with recyclers. This digital reconciliation prevents fraud and ensures deposit refunds are correctly applied. Retailers thus participate as data providers and gain streamlined accountability, while consumers get reliable deposit refunds, closing the loop.
- **Regulators and National Institutions:** The National Solid Waste Authority (ANDRES) and the Ministry of Environment (MARN) are primary users of the platform's reporting outputs. They gain **audit-ready, granular data on nationwide recovery volumes**. Instead of chasing paper reports, regulators can trust the on-chain evidence of compliance. They can set benchmarks (e.g. annual plastic collection targets) and transparently verify achievements or penalties. This alignment with regulatory needs makes enforcement efficient and fair.
- **International Donors and Climate Finance:** Although indirect users, development agencies (UNDP, GEF, climate funds) and global ESG investors form a "market" for verified environmental outcomes. By quantifying plastic recovery in a transparent ledger, we can support claims for carbon credits or other eco-credits. These stakeholders value the high-integrity data, which can attract new funding for expanded recycling.
- **Cardano Ecosystem and Tech Developers:** Community developers and blockchain enthusiasts form an ancillary user base. They interact with the open-source codebase, helping adapt and improve the platform. More importantly, they boost volume through system testing and can innovate additional features (like integrating IoT sensors or DPP systems). Their involvement helps propagate Cardano as the sustainability blockchain.

Each of these user profiles interacts with the platform differently. For example, recyclers and PROs engage through mobile or desktop apps to submit evidence, while regulators consume dashboards and reports. This diversity requires careful user experience design – simple reporting forms for collectors, comprehensive analytics for officials, and integration APIs for corporate users. By segmenting the market this way, we ensure the solution meets real-world needs across the value chain, from the informal worker on the ground to the policymakers at the top.

4. Product Description and Value Proposition

Our product is a blockchain-based platform that verifies and traces plastic recovery from collection to recycling, turning these activities into auditable digital assets. It is designed as a Web2-friendly interface with Web3 integrity: users easily upload data (e.g. weights, photos, GPS stamps) through intuitive screens or even SMS, and in the background every event is anchored on the Cardano blockchain. Key components include:

- **Digital Certificate Generation:** Every recorded recovery is verified (via automated checks) and converted into a digital certificate or token on Cardano. These certificates serve as **unforgeable "proof of recycling"**. They can be accumulated and burned to represent recycled plastic quotas, or swapped as **plastic credits** in emerging environmental markets.
- **Integrated MRV (Measurement, Reporting, Verification):** The platform replaces scattered spreadsheets with a standardized workflow. Collection points can weigh and photograph material, which is immediately hashed and time-stamped on the chain. Anomalies (e.g. duplicate entries, unrealistic weights) are automatically flagged for review. This ensures **high-quality, trusted data** every step of the way.
- **Automated Incentive Distribution:** Verified recyclers and collection centers receive timely payouts via smart contracts. Incentives might come from EPR fees, donor funds, or internal project budgets. Crucially, **payments are transparent and immediate**, relieving small operators from long cash-out delays. Each transaction is logged on-chain for accountability, and tokens are retired once claimed to prevent double counting.

- **Deposit Return (DRS) Support:** The architecture is DRS-ready. For pilot purposes, the system handles aggregated redemption batches for selected SKUs. Retailers input the number of bottles returned and deposits paid out; the platform reconciles this against distribution data from bottlers. This feature means El Salvador can **pilot deposit schemes at scale** without blockchain overload, validating returns in bulk rather than item-by-item.
- **Public Transparency Dashboards:** We deliver live dashboards and reports tailored to different audiences. A public-facing portal shows progress (e.g. total kilograms recovered, recycling rates) to build trust and prevent greenwashing. Private dashboards allow companies and regulators to drill into details (e.g. by region or SKU). All reports align with the data formats required by ANDRES and the Environment Ministry, so there is no extra translation work.
- **Accessibility and Inclusion:** Recognizing local conditions, the platform offers **offline functionality and feature-phone compatibility**. Workers can record data even in areas without internet and sync later. Field agents ensure that accreditation includes those with limited tech skills. This design broadens participation well beyond tech-savvy users, truly democratizing the system.

The **value proposition** is multifold:

- **For Grassroots Workers:** The platform offers dignity and reliability. Recyclers become part of a formal chain, earning verified income and gaining skills. We provide safety training, PPE guidance, and even community benefits (for example, medical check-ups during workshops) to uplift vulnerable groups.
- **For Businesses:** It brings **transparency and anti-greenwashing assurance**. Companies using the service can credibly claim "X kg of recycled plastic," backed by blockchain records. This bolsters their ESG credentials and prepares them for future regulations like Digital Product Passports. The platform can even convert certificates into plastic credits or tradeable tokens, creating new revenue or offset opportunities.
- **For Government:** The value is institutional efficiency and better outcomes. Regulators get a **one-stop, tamper-proof log** of all recovery activities, enabling them to enforce the waste law, issue fines or credits correctly, and

plan evidence-based policies. They can also report progress to international bodies (UNDP, UNEP) with confidence in the data.

- **For the Environment and Society:** Better traceability directly translates into more plastic being recovered and recycled, rather than leaking into the environment. This reduces greenhouse gas emissions from producing virgin plastic and prevents ocean pollution. By formally involving the informal sector, the platform also stimulates green jobs and inclusivity, aligning with UNDP's goals of no one left behind.

In essence, our solution makes plastic waste recovery credible, traceable, and economically valuable. It anchors intangible environmental actions in a public ledger, turning them into tangible impact for every stakeholder.

5. Branding and Positioning Strategy

The initiative will be branded as a **trusted partnership** between Plastiks and UNDP – leveraging Plastiks' reputation as an ISO-certified, blockchain-savvy innovator in verified recycling, and UNDP's status as a development leader. Our messaging emphasizes **"Verified Plastic Recovery for All"**, portraying the platform as the go-to digital infrastructure for a circular economy in El Salvador. Key positioning elements include:

- **Trust and Legitimacy:** As a UNDP-backed project using Cardano, the platform stands out from purely commercial solutions. Our branding highlights endorsements from government bodies (ANDRES, MARN) and partnerships with respected NGOs (FUNDEMAS). By showing the UNDP logo and Cardano Foundation affiliation, we instill confidence in funders, partners, and users.
- **Inclusion and Empowerment:** Marketing materials (flyers, posters, online campaigns) will feature local recyclers and community cooperatives, reinforcing that this technology is for them. Emphasis on words like "empowerment," "community," and "sustainability" align with UNDP's social impact narrative. Outreach programs will celebrate success stories (for instance, a woman cooperative that significantly increased earnings thanks to the platform), building a human face for the brand.

- **Compliance and Innovation:** To attract businesses, we stress "compliance-grade data" and cutting-edge infrastructure. Phrases like "Audit-ready ESG data" or "Blockchain-backed EPR compliance" will resonate with corporate sustainability officers. We position Cardano as the low-carbon blockchain choice, aligning with climate-conscious values. Visual branding will combine eco-themes (recycling symbols, clean environment) with tech motifs (blockchain icons, digital networks) to appeal across sectors.
- **Local Adaptation:** Although backed by an international solution provider, the platform will carry a **local identity**. It will use Spanish and possibly indigenous design elements to feel homegrown. We might adopt a project name in Spanish (e.g. "Rastreo de Plástico Verde") for national rollout, while retaining "Plastiks" as a known brand in corporate communications.
- **Channels and Community Engagement:** We will launch campaigns via UNDP El Salvador's networks, local media, and community radio to reach diverse audiences. Social media postings and events will include hashtags like #ElSalvadorCircular and #InclusiónPlástica. Attending environmental fairs and waste management conferences will raise visibility. In the Cardano community, co-marketing (as indicated in our Catalyst plan) will showcase El Salvador as a case study in sustainable blockchain use, aligning with Cardano's climate and inclusion mission.

By combining the reputations of Plastiks, UNDP, and Cardano, and tailoring the message to each audience (from informal workers to regulators to global observers), we establish the platform as the authoritative solution for waste traceability. The brand promise: **"Transforming recovered plastic into trusted, verifiable impact"**.

6. User Acquisition and Growth Plan

To achieve critical mass, our strategy focuses on **partnerships, capacity building, and incremental scaling**:

- **Engage with Local Hubs (FUNDEMAS and Cooperatives):** FUNDEMAS will lead a "train-the-trainer" program in collaboration with local recyclers' associations. We will organize introductory workshops and hands-on sessions in key municipalities (e.g. San Salvador and coastal communities where plastic pollution is acute). In these sessions, waste cooperatives learn to use the mobile app and see immediate benefits (for example, demo payouts). This seeding phase targets 8–10 pilot organizations in the first 4 months. Incentives such as starter kits (basic PPE, scales) are provided to encourage early participation.
- **Leverage UNDP Networks:** UNDP El Salvador's existing connections with municipal governments and community groups will be harnessed. We will integrate the platform into relevant public programs (like community waste cleanups) by having local officials endorse its use. Testimonials from respected figures (e.g. municipal mayors or community leaders) will motivate others to join. UNDP can also showcase the project at international development forums, potentially attracting funding or partnership commitments.
- **Partner with Producer and Retail Associations:** We will approach associations of beverage companies and retailers, framing the platform as a solution to upcoming EPR reporting requirements. Demo sessions for corporate sustainability teams will highlight how participation streamlines their compliance. Some early-adopter firms may sponsor parts of the system (e.g. cover operational costs or underwrite plastic credits) in exchange for branding and data access. This co-financing model accelerates growth and signals market demand.
- **Iterative Onboarding and Expansion:** After the initial cohort is accredited, a second wave of 30-50 cooperatives and operators will be recruited, using the first group as champions. Success stories (e.g. "Local cooperative earned X% more income with our platform") will be shared in local media and newsletters. We will also expand geographically, adding new municipalities every quarter based on plastic waste volume and community interest. By month 12, we aim to have trained and onboarded ~100 organizations nationwide.

- **Technical Enablement:** As more users join, we will deploy field support teams to assist with on-site setup and troubleshooting. Our platform includes a referral program: existing users can invite peers and receive a small token reward when the new user submits verifiable data. This creates organic growth among tightly-knit informal networks.
- **Feedback-Driven Iteration:** Early user feedback (gathered through surveys and field visits) will inform feature improvements – for example, adding regional language support or simplifying workflows. Demonstrating responsiveness builds trust and sticksiness.
- **Community Events and Storytelling:** Periodic events (like "Plastic Recovery Hackathons" or clean-up drives) will involve recyclers, students, and CSR teams, showcasing the platform in action. Social media will feature weekly updates on volumes recovered and positive impacts. By narrating the journey (e.g. "This week we reached 10,000 kg of plastic tracked!"), we maintain momentum and interest.

This multi-pronged approach ensures a steady ramp-up of users. The goal is to reach a tipping point where **the platform's value becomes self-evident** and word-of-mouth accelerates uptake. By blending institutional outreach with grassroots empowerment, we create a virtuous cycle: as more recyclers join and more plastic is traced, more businesses and municipalities see the benefits and come on board.

7. Go-To-Market Strategy

Launching the platform in El Salvador involves coordinated steps across policy, technology, and community channels:

1. **Official Pilot Launch:** Kick off with a high-profile event co-hosted by UNDP and the Ministry of Environment. This secures political buy-in and media attention. Demonstrations will show real-time tracking of plastic through the app. Invited guests include government officials, industry leaders (bottling companies,

retailers), NGOs, and waste cooperatives. This signals a serious, solution-driven partnership.

2. **Regulatory Alignment:** Work closely with ANDRES and MARN to integrate the platform into regulatory workflows. For example, agree that certified recovery data from our system will be an accepted input for companies' EPR reports. This may involve formal MoUs or policy directives. Ensuring official recognition of the data is critical for market confidence.
3. **Pilot Zones:** Initially deploy in select municipalities chosen for high plastic production or existing recycling activities. These zones act as "proof points." The system will include localized branding (e.g. co-branded with the city government) to reinforce local ownership. As these pilots produce results, they will be publicized as case studies to attract additional regions.
4. **Private Anchor Partnerships:** Secure partnerships with at least one major beverage company and one retail chain to supply SKU sales data for the DRS feature and to fund initial recovery. These companies can sponsor rewards for returning items (enhancing consumer buy-in), and help demonstrate the end-to-end cycle from sale to verified recycling. Their involvement lends credibility and financial support.
5. **Media and Communications:** Run a multimedia campaign around launch, using press releases, radio spots, and social media. The narrative will stress collaboration, e.g. "UNDP and local tech join forces to fight plastic waste." Human interest angles (stories of a recycler family helped by the project) will broaden appeal. Alignment with global campaigns (like UN Plastic Pollution Day) will be used for additional coverage.
6. **Scaling Beyond Pilot:** After 6 months of pilot data collection and iteration, prepare for scale-up. This includes migrating improvements (from lessons learned) and scaling server capacity. We will draft a roadmap to extend coverage nationally, setting milestones (e.g. double the number of collection points every 3 months). To reach new users outside the initial circles, we will leverage networks of environmental NGOs and possibly grassroots government programs.
7. **Leveraging Catalyst and Cardano Community:** Though the primary market is local, we will also engage with the global Cardano community. This involves publishing progress reports on Cardano forums and potentially setting up a

decentralized financing model (e.g. a Cardano stake pool dedicated to funding environmental projects). Demonstrating real transactions helps position Cardano as the public infrastructure for climate action.

8. **Continuous Training:** As part of launch activities, establish permanent training hubs at FUNDEMAS offices and other partner locations. New entrants can learn the system regularly, ensuring growth is sustained. Materials (videos, manuals) will be available online, adapted to local contexts.

By combining top-down support and bottom-up engagement, the go-to-market strategy ensures the solution moves swiftly from pilot to mainstream. The emphasis is on creating compelling first examples of success that entice broader adoption. In all activities, the solution is portrayed not just as an app, but as the backbone of El Salvador's circular economy – a national digital utility rather than a stand-alone product.

8. Impact Model and Measurement

The expected impact spans environmental, social, and systemic dimensions, each tracked with quantitative and qualitative metrics:

- **Environmental Impact:** We will measure the total mass of plastic waste recorded and recycled through the platform. For example, by the end of Year 1 we target **verifying tens of thousands of tons** of plastic (with an initial pilot goal of ~25,000 tonnes as a baseline). This translates into reduced landfill use and lower greenhouse gas emissions (recycling plastic uses significantly less energy than producing new plastic). The platform's public dashboard will display cumulative recovered volume, reduction in plastic entering waterways, and improvements in recycling rates. Independent audits or academic studies may be commissioned to assess local pollution levels (pre- and post-project) to capture broader environmental effects.
- **Social Impact:** Key social metrics include the number of individuals trained and certified as recycling operators, the number of female workers engaged, and the change in income for participating cooperatives. We will track how

many formal jobs are created or supported (directly or via cooperatives), and monitor well-being indicators through partner NGOs (for example, surveys on income stability or access to healthcare among participants). Specific targets might include accrediting ~100 operators (both formal and informal) within a year, and ensuring at least 40% of them are women or marginalized individuals. Community stories and testimonials will provide qualitative evidence of empowerment and social cohesion.

- **Economic Impact:** On the economic side, we will measure the flow of funds into the system (in ADA or fiat), such as total incentives paid to recyclers and potential revenue generated by selling plastic credits. For companies, metrics include the amount of recycled content (kg) they can credibly claim or offset. We will compute the value of plastic credits issued and the number of businesses engaged in EPR compliance via the platform. In the medium term, if the marketplace for plastic credits develops, we will track trade volume and price signals as an indicator of the platform's economic viability.
- **Policy and Compliance Impact:** The system's data enables new KPIs for the government: percentage of obligated producers meeting recovery targets, reduction in uncollected waste, and frequency of reporting. Improvement in reporting completeness and timeliness will be monitored (e.g. 90% of operators submitting monthly reports within 10 days of the period's end, compared to a much lower baseline). By quantifying and visualizing these metrics, the project helps enforce the Special Waste Law and demonstrates compliance in a transparent manner.
- **Innovation and Systemic Impact:** The project's success will also be measured by its replicability and adoption beyond the pilot. Indicators include documentation of the platform for other countries (shared open-source templates, APIs), and the formation of new collaborations (e.g. with other UNDP country offices or environmental agencies). Within Cardano, the metric is the growth in active wallets and transactions tied to environmental use cases. In Year 1, we expect over 100,000 Cardano transactions from the El Salvador track alone, showing real-chain usage.
- **Alignment with UNDP and SDGs:** We will map each output to Sustainable Development Goals. For instance, SDG 12 (Responsible Consumption and Production) is advanced by tracking plastic recovery; SDG 13 (Climate Action)

by reducing emissions; SDG 1 and 8 (No Poverty, Decent Work) by improving livelihoods of recyclers; and SDG 17 (Partnerships) by the multi-stakeholder collaboration itself. Progress against UNDP's country-specific environment and inclusion objectives will be documented in a final report.

Data for these metrics will come directly from the platform's records (ons-chain evidence provides an auditable data source), supplemented by partner reports (for socioeconomic indicators). Regular monitoring and evaluation (M&E) sessions will review performance against targets. This rigorous impact model not only validates the project's success but also provides accountability to all stakeholders, from village cooperatives to UNDP funders.

9. Post-Acceleration Sustainability Plan

Ensuring the platform endures beyond the pilot phase is a core consideration. Our plan focuses on **institutional integration, revenue models, and scalability**:

- **Institutional Ownership:** We will work from the outset to embed the solution within El Salvador's waste governance. This means drafting a roadmap for handing over certain responsibilities to ANDRES or another national body. For example, after 12–18 months the government could take on hosting the data dashboards or co-funding maintenance. We will seek formal agreements that ANDRES and the Ministry of Environment recognize and budget for the platform as part of their official EPR program. By making it the accepted "single source of truth," its usage becomes a norm rather than a project.
- **Funding and Business Model:** Parallel to grants, we will explore revenue streams. One approach is a **subscription model** for large producers or recycling companies who pay a small fee to access premium data analytics and certification services. Another is to monetize the plastic credits – for instance, companies might pay to retire credits for corporate offsetting, with fees covering operational costs. We will also seek ongoing support from international funds aligned with climate goals (e.g. Green Climate Fund) by

showcasing the platform's impact. Importantly, all pricing will be transparent and fair, ensuring small cooperatives still see net gain.

- **Local Capacity Building:** A key part of sustainability is having trained local teams. We will establish a support team within Nozama Tech's local partner network (and train UNDP country staff) to handle day-to-day operations, troubleshoot issues, and continue user training. This local presence, combined with online resources (documentation, tutorials), ensures that the platform can evolve after Catalyst funding ends. Additionally, the platform's codebase and processes will be well-documented and partly open-source, inviting community contributions and easing technology transfer.
- **Scaling and Replication:** The joint platform for Armenia and El Salvador demonstrates how easily the model can be replicated. We will package the implementation guide and toolkits (data templates, APIs, UI components) so that other countries or regions can quickly launch their own traceability systems. This broader scope appeals to Cardano's global community and to UNDP's network of country offices. We anticipate that success in El Salvador will attract interest in similar pilots elsewhere, creating a self-sustaining ecosystem of projects using our platform.
- **Continuous Improvement and Revenue through Innovation:** The platform is not static. We envision future modules (like integrating digital product passports or satellite imagery analysis) that can be commercialized or supported by research grants. For example, once operational, we could develop a premium analytics module for corporates that forecasts recycling needs or suggests infrastructure investments. These innovations, funded by private partners or international grants, feed back into sustaining the core service.
- **Community Commitment:** Lastly, by demonstrating tangible benefits to communities, we create public demand for the platform's continuation. When recyclers and municipalities come to rely on the transparency and income it brings, they will advocate for keeping it alive. UNDP will highlight success metrics in their reporting, making a case for continued support from national budgets and international partners.

Through these measures—policy integration, diversified funding, local empowerment, and an emphasis on replication—we ensure that the project's momentum carries well beyond the accelerator phase. In essence, it transitions from a funded pilot into a self-propagating solution that can adapt and grow, maximizing long-term impact.

10. Risk and Mitigation Strategy

We have identified several key risks and formulated mitigation strategies to ensure project resilience:

- **Institutional and Political Risk:** Change in government priorities or turnover in key agencies could stall progress. *Mitigation*: Formalize partnerships early with MOUs outlining roles and responsibilities. Involve multiple stakeholders (ANDRES, MARN, UNDP) in governance (e.g. Steering Committee) so that institutional knowledge is preserved. Demonstrating early wins will also build political momentum and create non-partisan support.
- **Stakeholder Coordination Risk:** Complex collaborations (UNDP, government, NGOs, private sector) could suffer misalignment. *Mitigation*: Establish clear communication channels and governance frameworks from day one. The planned monthly Steering Committee and weekly Technical Working Group ensure timely decision-making. Shared documentation (accessible online) and joint training sessions foster a unified approach.
- **Adoption and Engagement Risk:** Grassroots recyclers might be hesitant to adopt new technology, especially if literacy or trust is low. *Mitigation*: We prioritize user-centric design: apps with local language support, offline features, and simple workflows. Initial training cohorts will include prominent local recyclers as "community ambassadors" who vouch for the system. Safety nets like initial incentive bonuses for first reports will help motivate early use. We also plan inclusive outreach, specifically inviting women's cooperatives and marginalized groups, with tailored support (such as on-site assistance and peer mentors).
- **Technical and Infrastructure Risk:** Potential issues include unreliable internet connectivity, data security concerns, or Cardano network downtimes. *Mitigation*: The platform's ability to work offline protects against spotty

connectivity. All data on the app is encrypted, and on-chain entries are public but anonymous. We will maintain a robust server infrastructure with backups. Regular security audits (as ISO 9001 implies) and testing before major releases will minimize bugs. The Cardano blockchain itself has proven stability, but in rare cases of network issues we have fallback procedures to queue transactions and process them when the network is restored.

- **Data Integrity and Fraud Risk:** There is a risk of fake or duplicate data entries (e.g. a collector attempting to report the same kilos twice). *Mitigation*: Multiple layers of verification are built in: photo and GPS stamps, hashing/deduplication algorithms, and human review of flagged entries. Any suspicious pattern triggers an alert. Because blockchain records are immutable, any tampering attempts leave a trace. Regular audits will compare physical collection logs (from cooperatives) with on-chain data to detect inconsistencies.
- **Financial Sustainability Risk:** Reliance on grant funding can be unsustainable if not diversified. *Mitigation*: As described, we are developing a phased revenue strategy (service fees, credit sales) early in the project. We will also maintain lean operations where possible and explore co-funding opportunities with corporate sponsors (who may provide in-kind support like equipment or training). Transparent accounting will show funders the cost-effectiveness, making it easier to secure follow-on funding.
- **Regulatory Risk:** Future changes in waste management law or tax on blockchain transactions could affect operations. *Mitigation*: The solution is designed for regulatory agility. Since all data can be reconfigured (e.g. new fields added or business logic updated via smart contracts), we can quickly adapt to new rules. We will stay engaged with policymakers through UNDP, helping shape regulations that recognize and require blockchain-traceable data, turning a risk into a competitive advantage.
- **Social and Ethical Risk:** Issues like exclusion of certain groups, exploitation of workers, or data privacy breaches could arise. *Mitigation*: We have "inclusion by design": the project explicitly includes informal workers and has a grievance mechanism (contacts for complaints). Privacy is respected by storing personal details off-chain and using anonymized tokens. We will ensure fair compensation to workers and encourage non-harmful practices

(e.g. no children in hazardous recycling jobs). Health and safety will be integrated into training (for example, providing gloves and guidelines).

- **Technological Adoption Risk (within Cardano):** The Cardano ecosystem is evolving, and integration issues (e.g. wallet compatibility) could challenge users. *Mitigation*: We use widely supported wallet solutions and develop a simple in-app wallet for users. The tech team will monitor Cardano upgrades and work with community developers to ensure compatibility. By being early adopters, we also contribute feedback to improve the blockchain for social projects.

Through proactive governance, inclusive design, and technical safeguards, we have built robust mitigation plans. Periodic risk reviews will ensure that emerging issues are promptly addressed. Our track record (plausible from existing pilots) shows we can adapt—turning many potential risks into opportunities to strengthen the project. With these measures, the pilot is well-positioned to navigate uncertainties and deliver on its promise.